AMC 10 Topical Test 5

Worked solutions · Choose a question to review.

Question 1 · Answer D

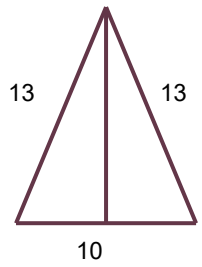
A 6 by 10 rectangle is cut into two equal rectangles by a straight cut parallel to its shorter sides. What is the sum of the two new perimeters?

A. 48 **B.** 38 **C.** 32 **D.** 44 **E.** 52

The original perimeter is 32. The 6-unit cut appears once on each new boundary, adding 12 for a total of 44.

Question 2 · Answer A

An isosceles triangle has sides 13,13,10. What is the distance from its apex to the midpoint of its base?



A. 12 **B.** 10 **C.** 5 **D.** 13 **E.** 8

The median from the apex bisects the base and is perpendicular to it. A right triangle has hypotenuse 13 and one leg 5, so the other leg is 12.

Question 3 · Answer C

A wheel of radius 6 rolls without slipping through three quarters of a revolution. The distance its centre travels is kπ. What is k?

A. 18 **B.** 12 **C.** 9 **D.** 6 **E.** 8

One revolution covers the circumference 12π. Three quarters covers 9π.

Question 4 · Answer C

On one straight line, B lies between A and C, and C lies between B and D. If AC=11, BD=13 and AD=19, what is BC?

A. 6 **B.** 4 **C.** 5 **D.** 8 **E.** 3

AC+BD counts the whole AD once and their overlap BC once more. Thus BC=11+13−19=5.

Question 5 · Answer B

An axis-parallel rectangle has three corners $(-2,3)$, $(5,3)$, $(5,-1)$. What is the square of its diagonal length?

A. 11 **B.** 65 **C.** 49 **D.** 33 **E.** 81

The horizontal difference is 7 and the vertical difference is 4. The Pythagorean theorem gives $49+16=65$.

Question 6 · Answer D

A walking track has two straight sides of length 20 and two semicircular ends of diameter 10. Its enclosed area is 200+kπ. Find k.

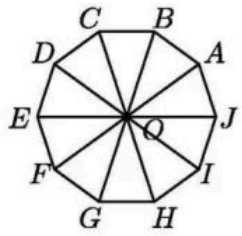
A. 50 **B.** 10 **C.** 5 **D.** 25 **E.** 100

The straight section is a 20 by 10 rectangle. Together the two semicircles make a radius-5 disk, contributing 25π.

Question 7 · Answer A

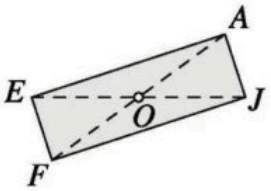
SOLUTION **A**

Let O be the centre of the regular decagon. The decagon is divided into ten congruent isosceles triangles by the radii joining O to each of the vertices of the decagon.The triangles AOJ and EOF are two of these ten congruent triangles.



Because O is the centre of the rectangle $AEFJ$ and the diagonals of a rectangle split its area into four equal areas, the triangles EOA and FOJ each have the same area as triangles AOJ and EOF.

Therefore the area of $AEFJ$ is equal to the area of 4 of the 10 congruent triangles that make up the decagon.



Hence the ratio of the area of $AEFJ$ to the area of the decagon $ABCDEFGHI$ is 4 : 10 which simplifies to 2 : 5.

FOR INVESTIGATION

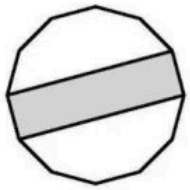
Prove the following geometrical facts that the above solution tacitly assumes. [A full solution would need to include these proofs.]

(a) The decagon has a centre, that is, there is circle which goes through all its vertices. (The centre of this circle is the centre of the decagon.)

(b) The centre of the regular decagon is also the centre of the rectangle $AEFJ$.

(c) The triangle EOA has the same area as the triangle AOJ.

The diagram shows a rectangle whose vertices are two pairs of adjacent vertices of a regular dodecagon.



What is the ratio of the area of the rectangle to the area of the dodecagon?

Generalize the result of the question and the above problem to the case of a rectangle whose vertices are two pairs of adjacent vertices of a regular polygon with $2n$ sides, where n is a positive integer.

The diagram shows a square inscribed in a regular dodecagon.



What is the ratio of the area of the square to the area of the dodecagon?

Question 8 · Answer D

In triangle PQR, a line joins the midpoints of PQ and PR. The perimeter of the smaller triangle at P is 18. What is the perimeter of PQR?

A. 54 **B.** 27 **C.** 24 **D.** 36 **E.** 30

The midpoint segment is parallel to QR, making a similar triangle with every side half as long. Double 18 to get 36.

Question 9 · Answer B

A square of area 18 is drawn with its diagonal horizontal. What is the distance between its two farthest vertices?

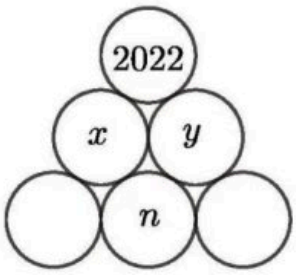
A. 12 **B.** 6 **C.** 9 **D.** 18 **E.** 3

In a square, diagonal squared is twice side squared. Since side squared is the area 18, the diagonal is $\sqrt{36}=6$.

Question 10 · Answer A

SOLUTION **A**

We let x and y be the positive integers in the discs in the middle row, as shown.



Since these are the products of the two numbers immediately below, n is a factor of both x and y. Hence n^2 is a factor of xy.

Now $xy = 2022$. It follows that n^2 is a factor of 2022.

The prime factorization of 2022 is

$$2022 = 2 \times 3 \times 337.$$

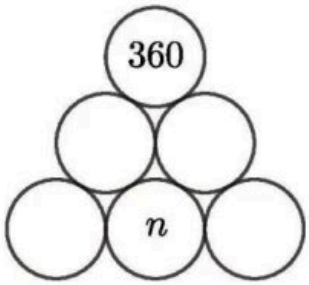
It follows that 1^2 is the only square of a positive integer that is a factor of 2022.

We deduce that $n = 1$.

FOR INVESTIGATION

- Check that $2 \times 3 \times 337$ is the factorization of 2022 into prime factors.
- In the number triangle shown on the right, each disc is to be filled with a positive integer. Each disc in the top or middle row contains the number which is the product of the two numbers immediately below.

  What are the possible values of n?



Question 11 · Answer C

A rectangular box has face areas 12,18 and 24 at one corner. What is the square of its volume?

A. 10368 **B.** 2592 **C.** 5184 **D.** 3888 **E.** 1296

If sides are a,b,c, multiplying ab,bc,ca gives $(abc)^2$. Thus the squared volume is $12\times18\times24=5184$.

Question 12 · Answer A

A right triangle has legs 9 and 12. What is the altitude from its right angle to its hypotenuse?

A. 36/5 **B.** 15/2 **C.** 9 **D.** 8 **E.** 6

The hypotenuse is 15. Both 9×12/2 and 15h/2 are the same area, so h=108/15=36/5.

Question 13 · Answer E

A rhombus has diagonals 10 and 24. A circle is tangent to all four sides. What is its radius?

A. 5 **B.** 120/13 **C.** 6 **D.** 4 **E.** 60/13

The half-diagonals form a 5–12–13 triangle, so the semiperimeter is 26. The area is 120 and equals inradius times semiperimeter. Hence r=60/13.

Question 14 · Answer C

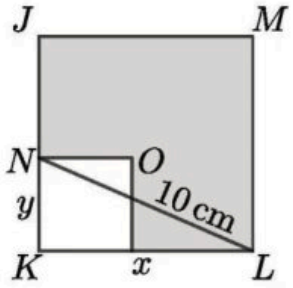
SOLUTION **C**

Suppose that the side length of the square $JKLM$ is x cm and the side length of the square $NKPO$ is y cm.

The area of the shaded region is the area of $JKLM$ minus the area of $NKPO$. Therefore

$$x^2 - y^2 = 62. \qquad (1)$$

Also, by Pythagoras' Theorem $KL^2 + NK^2 = NL^2$. That is,



$$x^2 + y^2 = 10^2. \qquad (2)$$

By subtracting equation (1) from equation (2), we obtain

$$2y^2 = 10^2 - 62 = 100 - 62 = 38.$$

Therefore $y^2 = 19$. Hence $y = \sqrt{19}$.

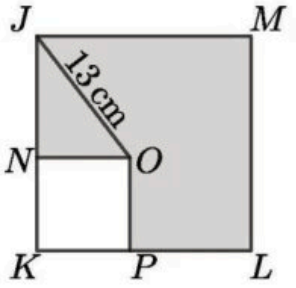
FOR INVESTIGATION

$JKLM$ and $KPON$ are squares.

The square $JKLM$ has area 289 cm^2.

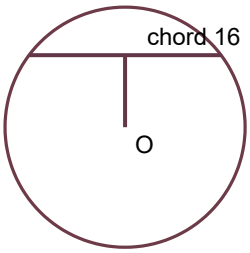
JO has length 13 cm.

What are the possibilities for the area of the square $KPON$?



Question 15 · Answer A

A chord 16 units long in a circle of radius 10 lies above its centre. How far from the centre is the midpoint of the chord?



A. 6 **B.** 8 **C.** 2 **D.** 4 **E.** 10

The radius to a chord midpoint is perpendicular. Half the chord is 8, so the distance is $\sqrt{(10^2-8^2)}=6$.

Question 16 · Answer D

A circle is tangent to both coordinate axes in the first quadrant and has centre on the line $x+y=14$. What is its diameter?

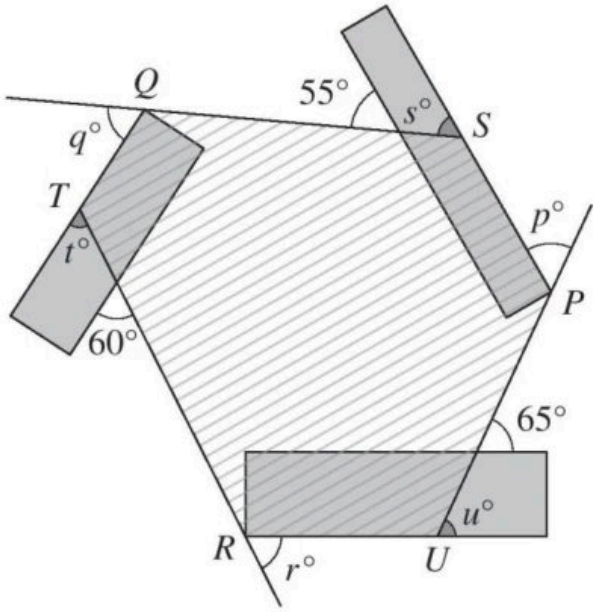
A. 12 **B.** 28 **C.** 7 **D.** 14 **E.** 21

Tangency to both axes puts the centre at (r,r). Therefore $2r=14$, already the diameter.

Question 17 · Answer B

SOLUTION **B**

Let P, Q and R be the points shown in the diagram where the rectangles touch the straight lines. Let the straight lines when extended meet the rectangles at the points S, T and U, as shown.



Then $PSQTRU$ is a hexagon. The external angles of this hexagon at the vertices P, Q and R are $p°$, $q°$, and $r°$, respectively, as given. Let the external angles at S, T and U be $s°$, $t°$ and $u°$, as shown.

The external angles of a hexagon have sum 360°. Therefore $p + q + r + s + t + u = 360$.

The opposite sides of a rectangle are parallel. When lines are parallel the corresponding angles are equal. Therefore $s = 55$, $t = 60$ and $u = 65$. Hence $s + t + u = 55 + 60 + 65 = 180$.

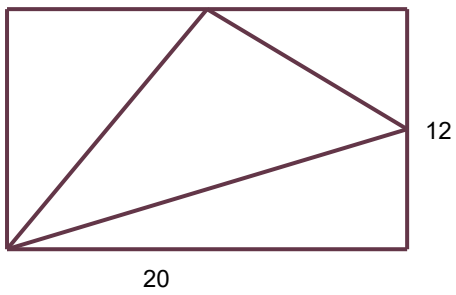
It follows that $p + q + r = (p + q + r + s + t + u) - (s + t + u) = 360 - 180 = 180$.

FOR INVESTIGATION

Explain why the sum of the external angles of a hexagon, and, indeed, of any polygon, is 360°.

Question 18 · Answer D

A 20 by 12 rectangle is divided by joining one corner to the midpoints of the two opposite sides. What is the area of the triangle with these three vertices?



A. 75 **B.** 60 **C.** 100 **D.** 90 **E.** 120

Take the corner at (0,0); the opposite midpoints are (20,6) and (10,12). Half the absolute determinant is (240−60)/2=90.

Question 19 · Answer B

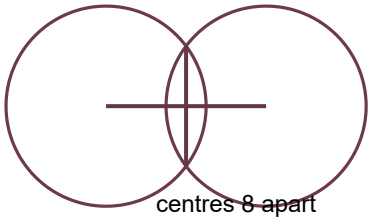
In triangle ABC, D is on BC with BD:DC=2:5. If the altitude from A to line BC is 8 and triangle ACD has area 60, what is the area of ABD?

A. 36 **B.** 24 **C.** 12 **D.** 28 **E.** 20

The triangles have the same altitude, so their areas are in the same 2:5 ratio as the bases. The smaller area is 60×2/5=24.

Question 20 · Answer C

Two equal circles of radius 5 have centres 8 units apart. What is the length of their common chord?



A. 4 **B.** 3 **C.** 6 **D.** 8 **E.** 10

The common chord bisects the line of centres perpendicularly. Its half-length is $\sqrt{(25-4^2)}=3$; hence its full length is 6.

Question 21 · Answer A

A right circular cone has radius 5 and slant height 13. Its volume is kπ. What is k?

A. 100 **B.** 300 **C.** 65 **D.** 125 **E.** 60

The axial right triangle gives vertical height 12. The volume is one third of π×25×12, giving 100π.

Question 22 · Answer B

A square has vertices (0,0),(8,0),(8,8),(0,8). A point P=(t,2) is equally distant from (0,0) and (8,8). What is t?

A. 5 **B.** 6 **C.** 7 **D.** 4 **E.** 2

Equate squared distances: $t^2+4=(8-t)^2+36$. Cancel t^2; $16t=96$, so $t=6$.

Question 23 · Answer E

From a point outside a circle a tangent has length 12. A secant from the same point has an exterior segment of 8. What is the length strictly inside the circle on that secant?

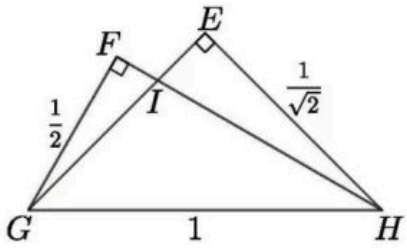
A. 12 **B.** 18 **C.** 8 **D.** 16 **E.** 10

Power of a point gives $12^2=8$ times the whole secant length, so the whole length is 18. Subtract 8 to get 10.

Question 24 · Answer D

SOLUTION **D**

We suppose that we have chosen units so that the length of GH is 1.



Because FGH is a $60°, 30°, 90°$ triangle, it follows that FG has length $\frac{1}{2}$.

Because EGH is a right-angled isosceles triangle it also follows that EH has length $\frac{1}{\sqrt{2}}$.

In the triangles IFG and IEH we have

$$\angle GFI = \angle HEI = 90°$$

and

$$\angle GIF = \angle HIE \text{ (vertically opposite angles).}$$

Because the sum of the angles in both these triangles is 180°, it follows that

$$\angle FGI = \angle EHI.$$

Therefore the triangles IFG and IEH are similar.

The ratio of the areas of similar triangles equals the ratio of the squares of the lengths of corresponding sides. Therefore

$$\begin{aligned} \text{area of } IFG : \text{area of } IEH &= FG^2 : EH^2 \\ &= \left(\tfrac{1}{2}\right)^2 : \left(\tfrac{1}{\sqrt{2}}\right)^2 \\ &= \tfrac{1}{4} : \tfrac{1}{2} \\ &= 1 : 2. \end{aligned}$$

FOR INVESTIGATION

- Explain why, given that GH has length 1, FG has length $\frac{1}{2}$ and EH has length $\frac{1}{\sqrt{2}}$.
- Explain why the ratio of the areas of similar triangles equals the ratio of the squares of the lengths of corresponding sides.
- (a) Given that GH has length 1, find the area of the triangle GIH.
  (b) Given that GH has length 1, find the areas of the triangles IFG and IEH.
  (c) Hence verify that the ratio of the areas of the triangles IFG and IEH is 1 : 2.

Question 25 · Answer D

A square’s vertices lie on a circle. Another square is made by joining the midpoints of the first. What fraction of the circle’s area is the smaller square’s area, expressed as k/π? Give k.

A. 2 **B.** 2/3 **C.** 1/2 **D.** 1 **E.** 3/2

If the circle radius is r, the first square has diagonal 2r and area $2r^2$. Its midpoint square has half that area, r^2. Dividing by πr^2 gives 1/π.